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Review article

The impact of intrinsic foot muscle strengthening exercises in asymptomatic athletes with normal foot structure: A systematic review

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ABSTRACT

Purpose: The intrinsic foot muscles (IFM) are crucial to the passive, active, and neurological subsystems that compose the foot core, and their weakness may predispose to various foot pathologies, systemic diseases and musculoskeletal conditions. Strengthening of IFM is an effective intervention in clinical conditions, but its role in sports and clinical rehabilitation is still inconclusive. Thus, this review aims to evaluate the effects of IFM strengthening exercises in athletes with normal feet.

Methods: An electronic database search was conducted using Cochrane Central Register of Controlled Trials (CENTRAL), Web of Science (WoS), Scopus, MEDLINE, PubMed, and the Physiotherapy Evidence Database (PEDro). The Cochrane risk of bias tool was used for assessing bias in the included studies.

Results: Four studies were included for qualitative review. The meta-analysis was not possible due to the heterogeneity of measuring units of outcome measures in the included studies. All the studies showed an improvement in intrinsic foot muscle strength, though the training protocols used by them were only roughly equivalent. One study reported no significant differences in MLA stiffness after 8 weeks of intervention. The clinician assessed motor performance and morphological assessments showed considerable improvements. Improved vertical running propulsive forces were reported, though the running economy showed no changes.

Conclusion: Findings suggest that IFM exercises effectively improve outcomes such as foot morphology and function, strength, running mechanics and subjective function in athletes and may be incorporated in sports rehabilitation. These conclusions should be considered with caution due to bias in included studies.

1. Introduction

The foot is a complex joint system with multiple degrees of freedom and plays a crucial role in athletic activities like running and sprinting.[1,2] The human foot has numerous intricate articulations and is essential for mobility. While producing the essential forces to support the human body, it also enables the execution of dynamic activities.[1] The feet operate as a propulsive lever and sustain the body's weight while continuously adjusting to the surface during walking, and they also help to disperse force. Any impairment of foot function puts the entire lower extremity's structural integrity at risk.[3] For healthcare providers to successfully treat any foot dysfunction, a deeper comprehension of normal foot function is necessary.

The "foot core" is made up of local and global stabilizers that support the foot skeleton similar to the lumbopelvic complex.[3] The primary stabilizers of the medial longitudinal arch (MLA), the intrinsic foot muscles (IFM), are crucial to the passive, active, and neurological subsystems that compose the core musculature of the foot. Additionally, they create the propulsion required for movement, provide muscular support to the base of support during postural stability, and help dissipate forces during loading.[4,5] The foot experiences greater stress than other parts of the lower limb, since it is the part of the lower limb closest to the external ground forces.[6] The IFM weakness is linked to several musculoskeletal conditions and systemic diseases,[7,8] and is also

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frequently accompanied by problems with balance, an increased navicular drop, decreased strength, and biomechanical anomalies that eventually lead to activity restriction and adversely impact patient function.[9,10] At every step, load redistribution may be more effectively managed by a stronger foot.[3] While decreased strength may make it more difficult to control inter-joint movements, which increases soft tissue strain, increased foot strength can potentially be an effective adaptation to the repetitive loading exerted on the foot while running, which could help to reduce the risk of injuries.[11]

Since humans are the only animals that must maintain their balance while running on single leg support, they differ from other animals (quadrupeds) in that they require a foot that is fairly mobile, able to adapt to uneven surfaces, and actively controlled.[3] Running or sprinting is a cyclic activity that involves all joints and muscle groups in the lower limbs, including the IFM and the metatarsophalangeal (MTP) joint. The MTP joint also plays a key role in sprinting, which may cause the population of sprinters to acquire unusually strong foot muscles.[12,13] The IFM are activated in response to increased postural demands, such as transitioning from a double to a single limb stance.[14] According to Tanaka et al. (2019), although elite sprinters have thicker foot muscles than non-sprinters, their Abductor Hallucis (AbH) thickness positively correlates with their 100-m personal best.[15] On the contrary, Yuasa et al. (2018) demonstrated a strong link between the ability to change direction in the pro-agility and 3-cone tests and maximal toe flexor strength with a dorsiflexed MTP in collegiate American football players.[16] Therefore, it is of interest to evaluate how these muscles work during running and to discuss possible effects of an IFM dysfunction for prevention or rehabilitation.[11]

A critical component lacking in the treatment of some overuse lower limb conditions could be the training of the intrinsic foot muscles to promote improved foot control. By enhancing the functional variability of degrees of freedom to handle the altering demands of dynamic foot control, training the intrinsic foot muscles may assist the foot core system.[11] For treating various foot and ankle problems, including chronic ankle instability, first MTP joint arthrodesis, patellofemoral pain, pes planus, and plantar fasciitis, IFM strengthening has been advocated.[7,8,17,18] Previous researches have demonstrated IFM strengthening exercise is effective in improving functional outcomes, including balance, strength, sensory function, motor performance, and subjective measures of pain and impairment in clinical groups in lower extremity injury rehabilitation protocols.[19–21]

The evidence establishing that IFM exercises increase functional outcomes in athletes with non-foot deformities, however, has not been thoroughly researched. It is important to note that researchers, trainers, and athletes are all working to enhance the athletes' performance in both training and competition. Therefore, any performance improvement in an athletic setting, regardless of its slightness, is considered significant. Despite the promising findings, it is essential to critically evaluate the existing evidence and identify any limitations or gaps in knowledge. There has been no systematic review and meta-analysis done till date that highlights the importance of strengthening the IFM in healthy athletes with non-foot deformities. By conducting a systematic review, this study aims to provide a comprehensive synthesis of the available research on the effects of intrinsic foot muscle-strengthening exercises in healthy athletes with normal foot structure. This review aims to find if IFM strengthening protocols augment athletes' performance and if so, how can it be incorporated in their training to act as a potential ergogenic aid. This review also aims to highlight the research gaps in terms of athletic training – the kinds of IFM strengthening protocols available, and methods to assess pre and post training strength gains of IFM. Also, reviewing the strengthening protocols for IFM in healthy athletes with non-foot deformities shall help researchers and professionals working with athletes comprehend baseline values of IFM strength gains, and these baseline values could then be extended as goals/desired results in other clinical groups in the athletic population. By consolidating the available evidence, this systematic review will provide a valuable resource for athletes and professionals seeking to optimize foot function and improve athletic outcomes.

2. Methodology

2.1. Registration

The International Prospective Register of Systematic Reviews (PROSPERO) lists the current review as CRD42023397869. This systematic review was reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) criteria.[22]

2.2. Study eligibility

This study includes control/clinical trials examining the benefits of strengthening IFM in healthy athletes (both recreational and competitive) with no foot pathologies or deformities. The clinical trials included in this study fulfilled the following requirements: Athletes (both recreational and competitive) with normal foot-structure. It also evaluated at least one parameter associated with sports performance, such as foot mobility, running biomechanics and foot motor performance tests. Studies were excluded if they involved an elderly population, foot deformity, recent injury and post-surgical cases, grey literature, and PEDro (Physiotherapy Evidence Database) scores below 5. The primary outcomes of this systematic study were in the areas of IFM strength, muscle activity, gait and/or running biomechanics, foot morphology and function.

2.3. Search strategies

The review question is "What are the effects of IFM strengthening exercises in healthy athletes with normal foot structure? It was formulated according to the PICO (Population, Intervention, Comparison, Outcome measure) framework. In the present review, the population is healthy athletes, the intervention is IFM strengthening exercises, the comparison is the control or alternate treatment group and outcome measures are IFM-related outcome measures in sports. The electronic databases included for performing the search were PubMed, Scopus, Web of Science, Central Register of Controlled Trials, and PEDro. The following keywords related to (intrinsic foot muscles OR intrinsic foot musculature OR intrinsic foot flexor OR toe flexors OR foot muscles OR foot core), (strength OR strengthening OR training OR therapy OR toe curls OR short foot exercises OR foot mobility OR muscle activity OR sprint OR sprint performance OR jump OR jump performance OR hop OR agility OR agility performance OR speed OR sport specific performance OR athletic performance) were used. To identify more articles relevant to the research issue, all of the aforementioned key terms were initially independently searched for and then combined. The search was restricted to English-language papers having the key phrases in their titles or abstracts.

Appropriate changes were made under each database's search system. The search items were verified for each database and all the retrieved studies were imported to the reference manager (EndNote™ version 21, Clarivate Analytics) for results synthesis and duplicate articles elimination. The title and abstract were used to group the remaining papers. After determining which full-text publications were appropriate, the final choices for data synthesis were made. Studies that satisfied all inclusion requirements were chosen. The flowchart for review selection is provided (Fig. 1).

2.4. Data extraction

Data were independently extracted from selected papers by three researchers. The extracted data included the author, year of publication, research design, participant details, characteristics of participants, the intervention provided, outcomes, and results. The data extraction was

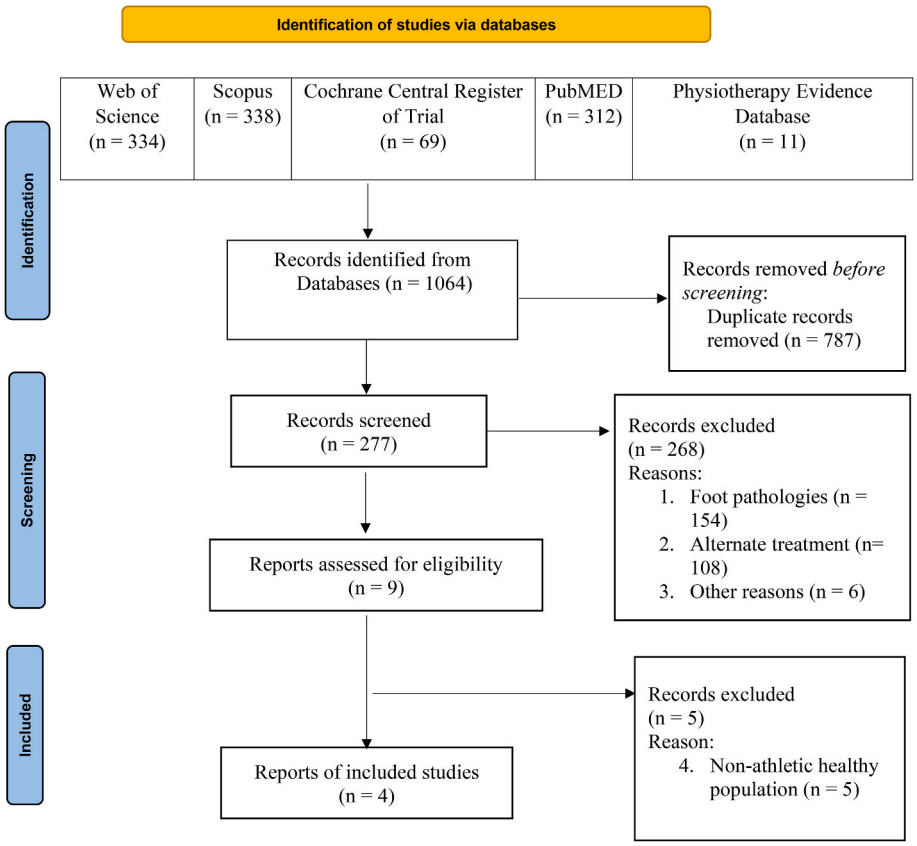

Fig. 1. PRISMA flow diagram, for eligible studies.

verified by another independent researcher, who also addressed any disagreements between the three researchers. The finalized data was examined in a group discussion and summarized and tabulated.

2.5. Risk of bias assessment

The risk of bias in the included studies was evaluated using the Cochrane Risk of Bias tool,[23] using the Review Manager (Revman 5.4, The Cochrane Collaboration). To identify the risk of bias in the included studies, the following seven domains were taken into account: 1) random sequence generation (selection bias), 2) allocation concealment (Selection bias), 3) blinding of participants and personnel (performance bias), 4) blinding of outcome assessment (detection bias), 5) incomplete outcome data (attrition bias), 6) selective reporting (reporting bias) and 7) other bias criteria were assessed (Fig. 2, Fig. 3).

3. Results

3.1. Search outcomes

Initial database search identified 1064 potential studies, and 277 abstracts remained after 787 duplicates were eliminated. Following a review of the abstracts, a preliminary evaluation was conducted to assess the eligibility of 9 full-text articles. After full-text analyses, four studies were deemed eligible for inclusion in the systematic review (Fig. 1).

3.2. Study characteristics

3.2.1. IFM strengthening setting

The basic characteristics of the four selected studies are shown in Table 1. The total number of participants in the four studies was 100, of which 55 were males and 45 were females. The intervention period was between 4 and 10 weeks, and interventions were given from 3 to 7 days per week. The total number of repetitions could not be calculated due to the varying number of frequencies of IFM strengthening exercises in a day.

3.2.2. IFM strengthening protocols

Although the protocols employed to strengthen the intrinsic foot muscles were not the same, their functions were fairly similar. Three studies have included short foot exercises,[24–26] while one study included Vele's Forward Lean and Reverse Tandem Gait exercises to strengthen the IFM.[27] E. M. Day and M. E. Hahn, (2019) adopted a dynamic exercise training protocol that incorporated isometric, concentric, and eccentric exercises which consisted of the range of motion, foot curls, eccentric and concentric band curls.[26] U. T. Taddei et al. (2020) included strengthening the plantar flexors, evertors and invertors along with the intrinsic foot muscles in their training protocol,[25] whilst Fraser and Hertel, 2019 focused only on the IFM exercises that included toe spread outs, hallux extensions and lesser toes extensions.[24]

3.3. Outcome measures

3.3.1. Strength measures

Two studies directly measured the hallux and toe flexor strength through a dynamometer and force platform.[25, 26]The other two studies reported strength measures through indirect assessments, with one using the Functional Movement Screen (FMS) Test.[27] One included study incorporated clinician-assessed motor performance and participant-perceived difficulty/Likert scales to report the same.[24]

3.3.2. Gait and running mechanics

U. T. Taddei et al. (2020) measured the vertical and anteroposterior

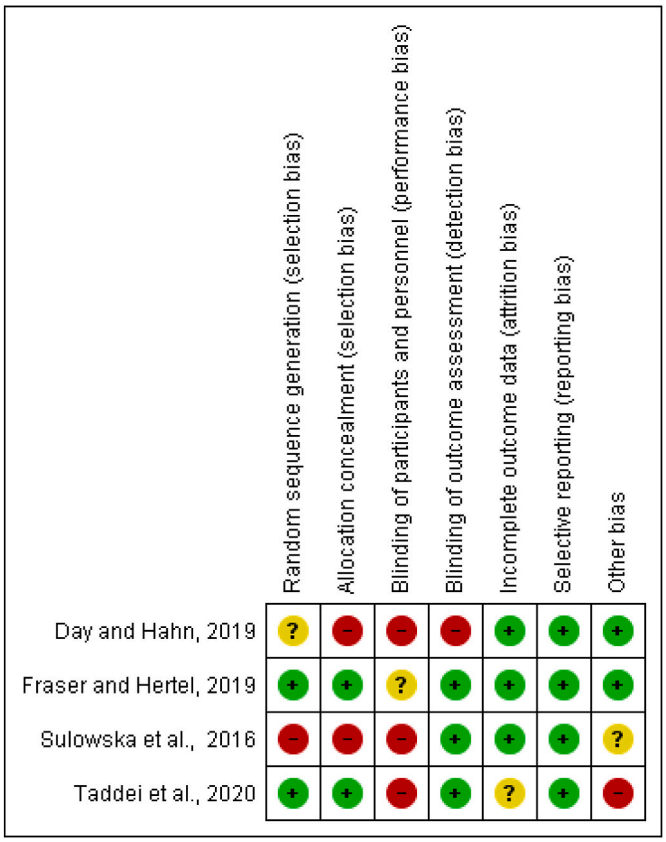

Fig. 2. Risk of bias summary: review authors' judgements about each risk of bias item for each included study.

propulsive impulses during running,[25] whereas E. M. Day and M. E. Hahn, (2019) assessed the gait biomechanics and running economy. The other two studies did not assess any such outcomes in their studies.[24,27]

3.3.3. Other outcomes

The outcomes for conditions other than IFM strengthening were IFM function and morphology, foot morphology, and the MLA range of motion (ROM) and stiffness. Two studies measured the IFM function and morphology. One included study measured the IFM anatomical cross-sectional area and volume through MRI and reported improved results for the same.[25] One included study) assessed motor activation of the AbH, Flexor Digitorum Brevis, Quadratus Plantae, Flexor Hallucis Brevis through ultrasound imaging and reported no significant changes.[24] While one included study reported no significant differences in MLA stiffness after eight weeks of intervention when assessing the ROM of the MLA during the stance phase of running and MLA stiffness during running.[24] In regards to foot morphology, one study showed significant improvement was observed in Foot Posture Index −6,[27] but another study reported no significant effects of the intervention on the Foot Health Status Questionnaire (FHSQ) score of any domain after eight weeks.[25]

3.4. Risk of bias

A grading of low risk of bias denoted that the study had specified the randomization process, deviations from the intended intervention, missing outcomes, outcome measures, or selection of outcomes to report, whereas a high risk of bias denoted that the study did not describe any of the aforementioned notions. A study categorized as "unclear" risk of bias meant that it was missing information, making it difficult to assess limitations and potential problems.

3.4.1. Sequence generation

Three of the four included studies were randomized control trials,[24–26] the fourth being a non-randomized clinical trial.[27] Two of the randomized control trials mentioned the method used by them to randomize participants, and even mentioned the allocation concealment done with the help of sealed envelopes, so these trials were considered at low risk of bias.[24,25] The third RCT reported that the subjects were randomized, but did not state the method of allocation and allocation concealment and hence was marked unclear.[26] The non-randomized clinical trial study was at a high risk of bias for both allocation and allocation concealment.[27]

3.4.2. Blinding

3.4.2.1. Participants and personnel. Three studies were at a high risk of bias for blinding of participants and personnel. Taddei et al. (2020) and Sulowska et al. (2016) reported that all participants were informed about the protocol after which their informed consent was taken for the same.[25,27] Day and Hahn, (2019) did not report about the blinding of participants.[26] The fourth study was deemed unclear as it did not clearly mention blinding the participants.

3.4.2.2. Outcome assessment. Three studies were at a low risk of bias for blinding the assessors.[24–27] One study was at a high risk of bias for not reporting assessor blinding.[26]

3.4.3. Incomplete outcome data

None of the four studies mentioned an intention-to-treat analysis, but three of them were at low risk of bias as two reported less than 20 % dropout rate,[24,26] while the third mentioned of no dropouts.[27] One study was deemed unclear for risk of bias for the missing information.[25] Less

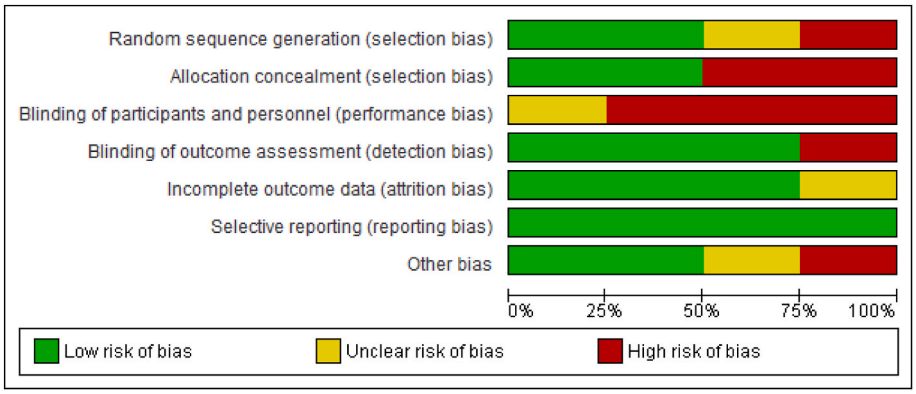

Fig. 3. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies.

Table 1
Table of eligible studies included in the review.

Authors	Characteristics of sample	Intervention duration	Type of intervention	Outcomes measured	Key findings	Conclusion
Taddei, Ulisses T. et al., 2020	28 recreational long-distance runners. (M: 9; F: 5 -CG) (M: 5, F: 9 -IG)	- 8-weeks; with a physiotherapist once a week and atleast 3 times at home. −100% of adherence would be a total of 112 completed sessions (14 subjects attending all 8 sessions).	IG: 8-week foot strengthening training. CG: 8 week 5-min warm-up and full body muscle stretching protocol before or after each running session.	- hallux and toe isometric strength. - intrinsic foot muscle ACSA and volume. - MLA range of motion and stiffness - foot function scores-Foot Health Status Questionnaire. - propulsive impulses during running (vertical, antero-posterior).	- Increase in the IG for the volume of all muscles investigated (AbH - 22.3% ↑, ABV - 12.1% ↑, FDB - 8.8% ↑, FHB - 17.7% ↑) and for vertical propulsive impulse during running.	- The 8-week foot exercise protocol effectively increased intrinsic Foot muscle volume and improved vertical running propulsive forces. - This shows that intrinsic muscle strengthening affects running mechanics and suggests that it may improve running performance.
Day and Hahn, 2019	23 competitive distance runners (8 M,15 F).	Once a day, 3 days per week for 10 weeks.	EG: isometric, concentric, and eccentric exercises which consisted of short foot exercise, band curl, and foot curls, once per day. - CG: not prescribed any additional strengthening protocol to augment their current training.	- Toe-flexor strength. - gait mechanics - running economy.	- Toe-flexor strength increased in EG, however, MTPJ and ankle mechanics and RE did not change. (EG toe-flexor strength −16% > at week five, and 27% > at week ten)	While the results do not support the notion that increased toe-flexor strength may improve RE and subsequent distance running performance, further research may be warranted to investigate the interplay between foot strengthening and footwear construction, fatigue resistance, as well as injury prevention or rehabilitation.
Fraser and Hertel, 2019	Healthy, recreationally active individuals. (n = 24–12 M, 12 F; mean age 21.5 ± 4.8 years; BMI 23.5 ± 2.9)	3 times per week for 4 weeks. 104 reps in 12 sets.	- IG received a 4-week progressive home IFM exercise program performed daily. - CG did not receive any intervention.	- Clinician-assessed motor performance. - Participant-perceived difficulty. - USI motor activation measures of the AbH, FDB, QP, FHB.	- IG demonstrated significant improvement in motor performance in the toe-spread-out exercise and less perceived difficulty in the toe-spread-out, hallux-extension, and lesser-toe-extension exercises. - Both groups demonstrated increased USI motor activation of AbdH during the toe-spread-out exercise.	A 4-week IFM exercise intervention resulted in improved motor performance and decreased perceived difficulty when performing the exercises, but not changes in USI measures of IFM activation compared to CG.
Sulowska, Iwona et al., 2016	25 long distance runners (11 F and 14 M) (Mean age 28 ± 3.86)	Daily for 6 weeks. −2 times a day for 15 min each time.	G1: (n = 13) Vele's Forward Lean and Reverse Tandem Gait exercise. G 2: (n = 12) Short Foot Exercises.	- FPI -6. - FMS test.	- Significant improvement in FPI -6 (talar head palpation in G1, and calcaneal inversion/ eversion calcaneus in G2). - In G1 a significant improvement was noted in FMS tests: deep squat, active SLRs and in total score.	Both training programs are effective with slight superiority of Vele's Forward Lean and Reverse Tandem Gait.

M: Male; F: Female; IG: Intervention Group; CG: Control Group; EG; Experimental Group; ACSA: Anatomical Cross Sectional Area; MLA: Medial Longitudinal Area; MTPJ: Metatarsophalangeal Joint; RE: Running Economy; USI: Ultrasound Imaging; AbH: Abductor Hallucis; ABV: Abductor digiti minimi; FDB: Flexor Digitorum Brevis; QP: Quadratus Plantae; FHB; Flexor Hallucis Brevis; G1: Group 1; G2: Group 2; FPI-6: Foot Posture Index- 6; FMS: Functional Movement Screen; SLR: Straight Leg Raise.

than or equal to a 20% dropout rate is within the permissible range for taking into account a low attrition bias.[28] Therefore, we classify dropout rates of 20% or above as having a high risk of bias, while dropout rates of less than 20% were deemed to have a low risk of bias (Figs. 2 and 3).

3.4.4. Selective reporting

All the included studies were at low risk of bias for reporting all the outcome measures clearly at pre and post the intervention.[24–27]

3.4.5. Other sources of bias

The included studies were deemed high risk of bias if they were not clinically registered and did not justify their limitations,[24–27] reported biasness while assessing the hallux and toe flexion strength using the pressure platforms, and hence was deemed at high risk of bias for the same.

3.5. Effects of intervention

U. T. Taddei et al. (2020) reported that the 8-week foot exercise

protocol effectively increased intrinsic foot muscle volume and improved vertical running propulsive forces,[25] whilst E. M. Day and M. E. Hahn, (2019) reported that toe-flexor strength increases in the experimental group did not change the MTPJ and ankle mechanics and running economy.[26] A 4-week IFM exercise intervention resulted in improved motor performance and decreased perceived difficulty when performing the exercises, but no changes were observed in USI measures of IFM activation compared to control group, as stated by Fraser and Hertel, (2019) and Sulowska et al. (2016), noted that both training programs targeted for IFM strengthening are effective in improving the FPI-6 and FMS scores, with slight superiority of Vele's Forward Lean and Reverse Tandem Gait.[24,27]

4. Discussion

4.1. Main findings

A systematic review was undertaken to determine the effects of IFM strengthening in healthy athletes with normal foot structure. All four studies reported improvement in IFM strength, with three studies reporting improvements in other outcome measures, whilst one study did not show any other gains other than muscle strength.

4.2. IFM strength

All the studies showed an improvement in strengthening the intrinsic foot muscles, even when the exercises were only slightly similar. The Short Foot Exercises (SFE) was the most commonly utilized IFM exercise in the studies that were included, except for one of the two groups of Sulowska et al., 2016, that employed the Vele's forward Lean exercise and reverse tandem gait as part of the training.[27] Most effective intervention to strengthen IFM and enhancing both subjective and objective outcomes seems to be SFE. The SFE is a maneuver wherein the midfoot is contracted and the metatarsal heads are moved posteriorly towards the calcaneus without toe flexion.[29] The medial longitudinal arch height is raised as the IFM is targeted for activation in this maneuver.[29] The IFM is hypothesized to be selectively activated by SFE while the extrinsic foot muscles remain inactive.[29] In all of the studies included in this review that used it, with the exception of one intervention group of Sulowska et al. (2016), it was shown to be more effective than control groups.[27]

The question of whether toe flexion or toe grip are reflective of similar muscle recruitment patterns or quite different is still unclear because toe strength testing procedures vary between studies.[30] The recruitment of extrinsic foot muscles (EFM; which have their bellies in the low limb rather than the foot), may be greater during toe grip than during toe plantar flexion, which favors intrinsic foot muscles.[31,32] Previous research indicates that, while the precise magnitude of the increases varied between muscles and locations, both the IFM and EFM muscles were typically more active during the single-limb positions as compared to the corresponding double-limb positions, and both the groups coordinated their activity together while moving from double-limb to single-limb tasks; hence indicating integrating IFM strengthening along with that of EFM would prove to be beneficial.[33] It is not apparent which motion, if any, provides a better representation of the muscle recruitment during stance and balance, and whether this is at all relevant.[30] The flexor digitorum brevis, lumbricals, interossei, flexor hallucis brevis, flexor hallucis longus, flexor digiti minimi brevis, flexor digitorum longus, and quadratus plantae are the muscles involved in toe flexion.[34] But differences in muscle origin and insertion locations of these muscles also have shown differences in their functions. For example, the flexor hallucis longus (FHL) and brevis (FHB) muscles both provide a plantar flexor torque at the MTP joint; additionally, the FHL muscle also produces plantar flexor torques at the ankle and IP joints due its anatomic location.[35] Thus, enhancing function during tasks involving a neutral or plantar flexed hallux may target the FHL along with other extrinsic foot muscles; whereas improved MTP joint torsional stiffness and MTP plantar flexion torque in the toes during different phases of propulsion during gait can be achieved by targeting the FHB and possibly the intrinsic foot muscles more generally.[36] Although it has been challenging to find evidence for this in the literature, it may be important to take into account in future research to develop a more precise understanding of foot muscle activity and activation patterns in postural and dynamic balance.

The IFMs have recently been the subject of multiple systematic reviews. Futrell et al. (2022) studied the effects of an intervention on functional mobility, while Willems et al. (2022) investigated the effects of a plantar IFM strengthening intervention on dynamic balance control and foot function during walking in adults.[9,37] Neither intervention approach, however, was restricted to performing SFEs. Different training protocols employed to strengthen the IFM indicate that no protocol is more advantageous than the other, as almost all the exercises are similar in function and work to train the IFM. Our study provided a review of studies reporting the effectiveness and interventions used to strengthen the IFM in healthy athletes with normal foot structure.

All the training protocols were longer in duration, with the minimum being 4 weeks, and maximum 10 weeks. The frequency and time duration varied for each protocol, with 2 studies training for three times per week,[24,26] one for atleast four times a week,[25] and one for all seven days a week.[27] Two studies directly measured the IFM strength and both reported an increase in the muscle strength.[25,26] The other two studies indirectly measured the muscle strength and reported in terms of various clinician assessed and patient reporting rating scales, and their results showed improvements in the scores of the rating scales.[24,27]

Two studies measured the IFM function and morphology. One study[25] measured the IFM anatomical cross-sectional area and volume through MRI and reported improved results for the same, whereas the other[24] assessed motor activation of the AbH, Flexor Digitorum Brevis, Quadratus Plantae, Flexor Hallucis Brevis through ultrasound imaging and reported no significant changes. One study reported no significant differences for MLA stiffness after 8 weeks of intervention when assessing the ROM of the MLA during the stance phase of running and MLA stiffness during running.[25] In regards to the foot morphology, one included study showed significant improvement was observed in Foot Posture Index −6.[27] While another included study reported no significant effects of the intervention on the FHSQ score of any domain after eight weeks.[25] Future research on examining strength training methods in enhancing the intrinsic foot musculature strength is essential, with a focus on healthy athletic populations and reliable outcome measurements.

4.3. Gait and running mechanics

Ulisses T Taddei et al. (2020) measured the vertical and antero-posterior propulsive impulses during running[25]; whereas E. M. Day & M. E. Hahn, (2019) assessed the gait biomechanics and running economy[26] found that the vertical running propulsive forces were improved by the 8-week intrinsic foot exercise protocol in recreational long-distance runners, however, the running economy did not change even with increases in the intrinsic foot muscle strength of competitive distance runners. The active and neural subsystems that compose the foot core comprise the IFM, which serve as the primary local stabilizers of the foot.[3]

These muscles, which have anatomical insertions under the foot, lengthen eccentrically during the early stance phase of running and produce negative work before shortening during the late stance phase as the arch recoils to produce positive work.[38] The following three muscle-tendon units: AbH, flexor digitorum brevis, and quadratus plantae facilitate this active contraction, which promotes propulsion.[39] One study demonstrated that IFM activity in late stance is required to produce enough impedance at the MTP joints, which in turn allows an efficient push-off.[24] Although both intrinsic and extrinsic foot muscles work together to provide toe flexor strength, the IFM appear to be the

primary sources of torque for MTP joints.[24]

High levels of strength in the small IFM are necessary for the biomechanical specificities at the forefoot and midfoot regions during running or sprinting.[2,40] In addition to handling significant amounts of constraints during the absorption phase, particularly impact attenuation, the foot core system must function as a strong and rigid lever to transfer lower limb forces as efficiently as possible during propulsion.[2] Deficits in active foot stabilization while running may be connected to overuse injuries involving control of the foot arch, which may result in higher tissue stresses.[2] Thus, an effective IFM strengthening protocol may help improve running performance, though more researches are required to understand the effects of these interventions and their efficient strengthening strategies.

Isometric exercises do not accurately represent the biomechanical function of the foot muscles during movement.[24] Running or even walking exerts a significant number of forces at the midfoot that it is impossible for the foot muscles to provide enough force for low load movements like the SFE. In order to get closer to the precise function of running, moving from isometric to plyometric activities may be beneficial, as seen in the studies by and suggested by.[2] They also suggested to take into account exercises that transfer the Center of Pressure forward considering they will impose more pressure on the midfoot.[2]

5. Limitations

In this systematic review, only English-language articles published in peer-reviewed journals were considered. The 4 studies included in this review were clinical trials only and had solicited significantly different IFM training protocols. The outcome is less likely to be consistent because there are no established protocols for enhancing athletic performance. Additionally, because the systematic review did not focus on a single sport, various sports may have varied regular training routines. These variables may also affect the outcome. RCT studies with a specific protocol and a larger sample size are necessary for a more thorough comprehension. Also, due to the heterogeneity in the data, meta-analysis was not feasible. Considering these limitations, the findings of the present review should thus be deliberated with caution.

6. Strengths

In defiance of the above limitations, this systematic review has promising findings. This is the first systematic review that reports of the effects of incorporating IFM strengthening exercises in athletic population with normal foot structure. The IFM are crucial to the foot core system, and their dysfunction may be a factor in predisposing to lower limb injuries. Also, the addition of intrinsic foot strengthening exercises in the training of athletes may help implore the effects on sports performance. Therefore, integrating strengthening exercises targeting the IFM may help improve sport specific performance and reduce the vulnerability to sports injuries. This review includes only randomized control trials, which are considered to be high level of evidence due to their reduced risk of systematic errors and biasness.

7. Future scopes and implications

Future research on examining strength training methods in enhancing the intrinsic foot musculature strength is essential, with a focus on healthy athletic populations and reliable outcome measurements, as there has been no standard protocol to train the same. The duration of the training interventions, strength assessment measures and sport specific measures also vary and hence there is a need to further improve the quality of interventions. We propose standardizing the methodological aspects of the subsequent IFM-strengthening studies by specifically addressing the doses, training interventions and assessment methods. The heterogeneity highlighted in this review amongst the researches also provides the researchers information about the research gaps present in the literature.

8. Conclusion

This systematic review has found evidences that indicate the efficacy of IFM exercises in a healthy sports population with normal foot structure that may be helpful to further contextualize the findings in clinical populations of interest and injury prevention strategies. The IFM exercises have been shown to be effective in improving outcomes such as foot morphology and function, strength, running mechanics and subjective function in studies involving healthy athletes. The results of these studies offer potential for understanding how the IFM exercises work. The IFM function, deficits, and treatments are still being explored in the literature. Incorporating the findings of the included research in their proper context can be crucial for clinical decision-making when treating lower extremity injuries and can enhance athletic performance by serving as an efficient ergogenic aid. Physiotherapists and other medical professionals, coaches and athletes may employ the existing medical and scientific literature to set up the exercises that are best suited to strengthening their feet, and further researches are warranted to set the variation and progression required and range from isometric, concentric to eccentric contraction modes, and from analytic to functional exercises.

Ethical approval statement

Given the nature of our manuscript as a systematic review, ethical approval from a specific board was not applicable. However, to ensure the rigor and transparency of our review process, we registered our review protocol with PROSPERO, the international prospective register of systematic reviews. Our registration number is CRD42023397869, which has been duly cited within the manuscript. This registration underscores our commitment to high ethical and methodological standards.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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